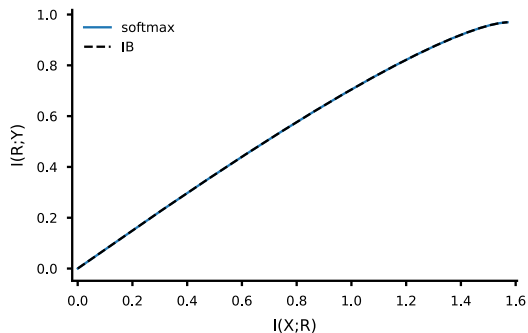


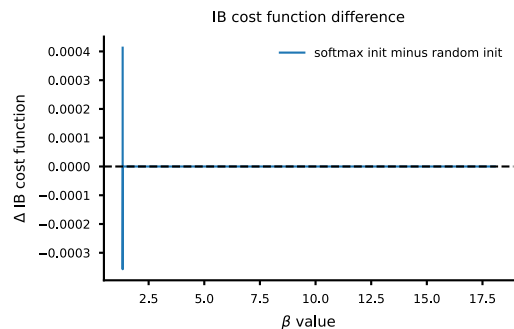
A

Three-jar bead-prediction task Fully-optimal (seven-back) IB bound



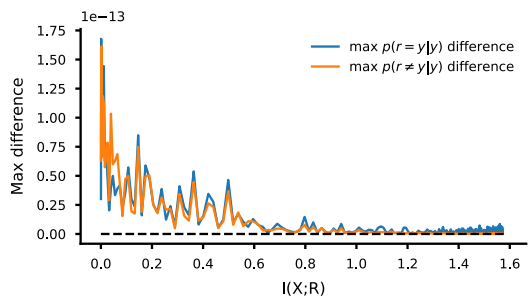
B

Comparison of softmax initialization to random initialization for IB algorithm (<0 means random is worse)



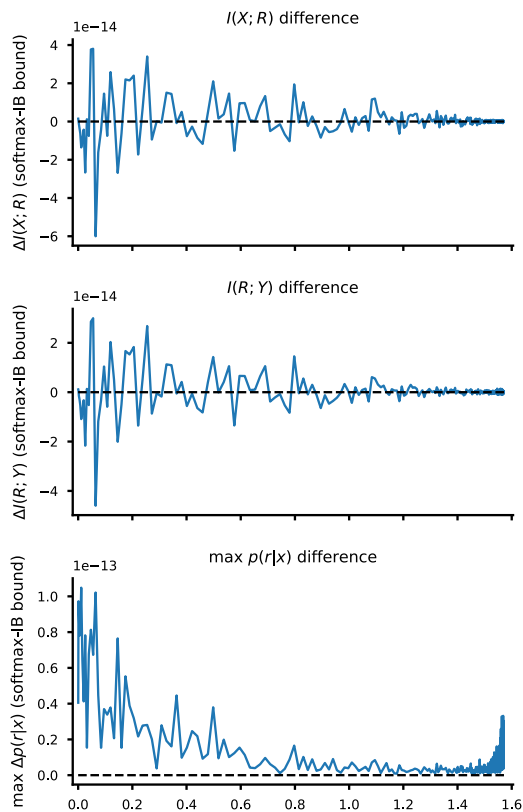
C

Check of $p(r=y|y)$ property that must be true for softmax solution to hold (traces must be 0)



D

Comparison of softmax solution to IB algorithm solution (0 means no difference)



E

Hidden Markov model parameters

Transition matrix, **H**
P(jar trial $t+1$ | jar trial t)

$$\begin{bmatrix} 0.98 & 0.01 & 0.01 \\ 0.01 & 0.98 & 0.01 \\ 0.01 & 0.01 & 0.98 \end{bmatrix}$$

Emission matrix, **E**
P(bead trial t | jar trial t)

$$\begin{bmatrix} 0.70 & 0.15 & 0.15 \\ 0.15 & 0.70 & 0.15 \\ 0.15 & 0.15 & 0.70 \end{bmatrix}$$